Li Sun

lisun-ai.github.io github.com/lisun-ai lisun@bu.edu (412)-209-5278

EDUCATION

Boston University *Aug 2024*
Ph.D. in Computer Engineering Boston, MA

SELECTED PUBLICATIONS

Sun, L., Luisier, F., Batmanghelich, K., Florencio, D., & Zhang, C. (2023). From Characters to Words: Hierarchical Pre-trained Language Model for Open-vocabulary Language Understanding. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. **Best Papers Honorable Mention**

Sun, L., Liu, H., Jia, S., He, Y., & You, C. (2025). DocAgent: An Agentic Framework for Multi-Modal Long-Context Document Understanding. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Sun, L., Ahuja, C., Chen, P., D'Zmura, M., Batmanghelich, K. & Bontrager, P. (2025). Multi-Modal Large Language Models are Effective Vision Learners. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

Robinson, J., **Sun, L.**, Yu, K., Batmanghelich, K., Jegelka, S., & Sra, S. (2021). Can contrastive learning avoid shortcut solutions? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Sun, L.#, Yu, K.#, & Batmanghelich, K. (2021). Context Matters: Graph-based Self-supervised Representation Learning for Medical Images. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

Saeedi, A., Utsumi, Y., **Sun, L.**, Batmanghelich, K. & Lehman L. (2022). Knowledge Distillation via Constrained Variational Inference. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

WORK EXPERIENCE

Amazon *July 2024 - Current*
Applied Scientist, AWS AI Seattle, WA

Customized Multi-Modal LLM Development

- Led supervised fine-tuning (SFT) and reinforcement learning (RL) post-training to develop customized LLMs, deployed in Fortune 500 customers.

Kiro Autonomous Agent

- Developed an LLM-powered virtual software developer that leverages organizational knowledge and codebase context to enable more personalized and accelerated development workflows.
- Designed and implemented an LLM agent for infrastructure-as-code generation, contributing to the product launch on Kiro.
- Filed a U.S. patent application as an inventor, and submitted a paper manuscript.

Meta *May 2023 - Sept 2023*
Research Scientist Intern, PyTorch Team New York, NY

- Proposed LLM-augmented visual representation learning (LMVR), leveraging multi-modal large language models to extract semantically rich, object-level image representations for downstream vision tasks.
- Developed token-level attention visualizations to interpret multi-modal LLM behavior and reveal word-to-image-region correspondence in generated visual descriptions.
- Evaluated LMVR across multiple datasets, demonstrating improved visual representation

generalizability under domain shift and enhanced robustness to corrupted and perturbed images.
- Manuscript accepted by WACV.

Microsoft *June 2022 - Aug 2022*
Research Intern, Microsoft Azure AI Bellevue, WA
- Designed and pretrained a hierarchical language model for open-vocabulary language understanding, replacing fixed sub-word vocabularies with a two-level intra-word and inter-word architecture.
- Experiments demonstrated that our model outperforms baselines, and it is more robust to textual corruptions and domain shifts.
- Manuscript accepted by ACL.

Microsoft *Mar 2019 - Aug 2019*
Research Intern, Microsoft Research Asia (MSRA) Beijing, China
- Developed fine-grained model that makes use of anatomical structure for chest radiograph interpretation.
- Achieved improved results on thoracic diseases that are subtle and require close observation.
- Developed reinforcement learning model for interpretable life support device detection.

SKILLS

Languages:	Python, C, C++, MATLAB, R, Java, JavaScript, SQL, HTML, CSS
Frameworks & Tools:	PyTorch, TensorFlow, LangChain, LangGraph, Git, Spark, Hadoop, Docker, Linux/Unix

SERVICE

Invited reviewer for NeurIPS, ICML, ICLR, CVPR, AAAI, IJCAI, MICCAI, IEEE TMI, IEEE J-BHI, etc.